



INVESTIGATION OF XXXXXXXXX

Student 1 Name SURNAME
Student 2 Name SURNAME
Student 3 Name SURNAME
Student 4 Name SURNAME
Student 5 Name SURNAME



Supervisor: Dr. Name SURNAME

ABSTRACT

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INTRODUCTION

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2009).

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THEORETICAL REVIEW

Time Evolution Equations:

• Franzetti(1982) $\frac{d_s}{d_{se}} = (1 - \exp(-BT^C))$

Where;
 d_s : scour depth T : dimensionless time

d_{se} : equilibrium scour depth t : time

B and C : constant t_e : equilibrium time

F_d : densimetric Froude Number density

σ_g : geometric Standart Deviation for sediment particles

t_R : reference Time

Maximum Scour Depth Clear Water Condition:

• Hancu(1971): $2.42 \left(\frac{y}{b}\right)^{1/3} Fr^{2/3}$

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Maximum Scour Depth for Live Bed Condition:

• Neil(1964)

$$\frac{d_{se}}{b} = 1.35 \left(\frac{y}{b}\right)^{0.3}$$

• Shen(1969)

$$\frac{d_{se}}{b} = 3.4F_r^{\frac{2}{3}} \left(\frac{y}{b}\right)^{1/3}$$

• Breusers at al.(1977)

$$\frac{d_{se}}{b} = 2 \tanh\left(\frac{y}{b}\right)$$

Where;

b : Pier Width

V : Approach Velocity

y : Approach flow depth

V_c : Critical Velocity

F_r : Froude Number

d_{50} : Median Grain Size

N_s : Number of Grain

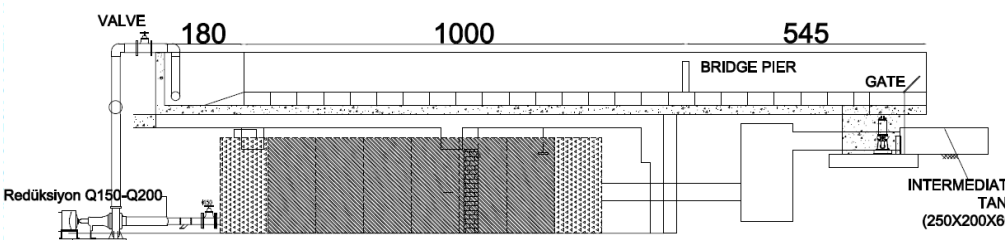
b^* : Effective Diameter

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PURPOSE OF THE STUDY

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EXPERIMENTAL SETUP

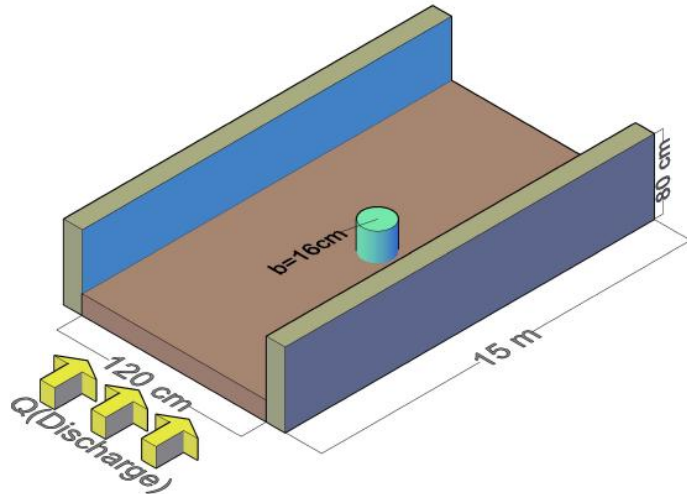


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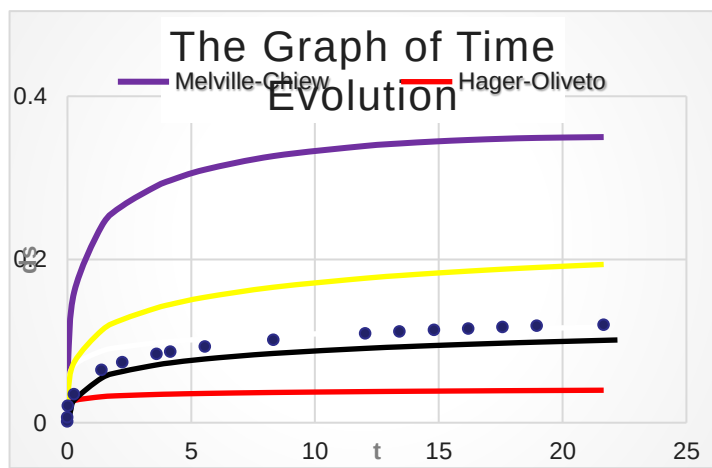


Scenario for Clear Water	Scenario for Live Bed
$\frac{V}{V_c} = 0.8$	$\frac{V}{V_c} = 1.2$
$Q=67$ l/s	$Q=100$ l/s

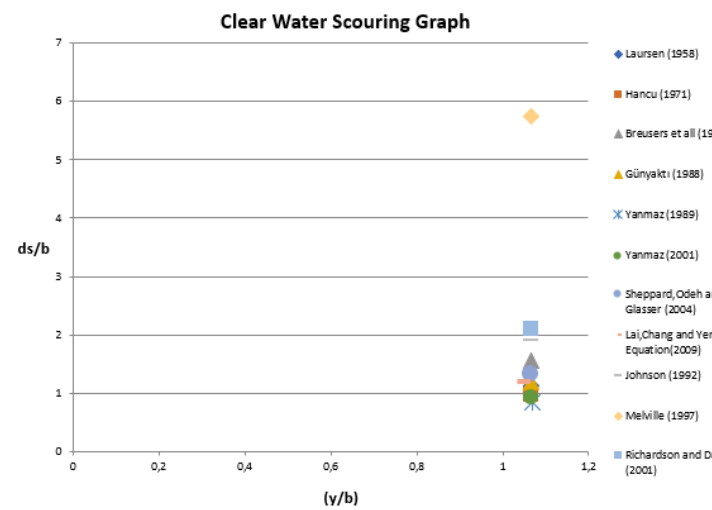
General Properties
$y=0.17$ m
$B=1.2$ m
$b=0.16$ m
$d_{50} = 1$ mm



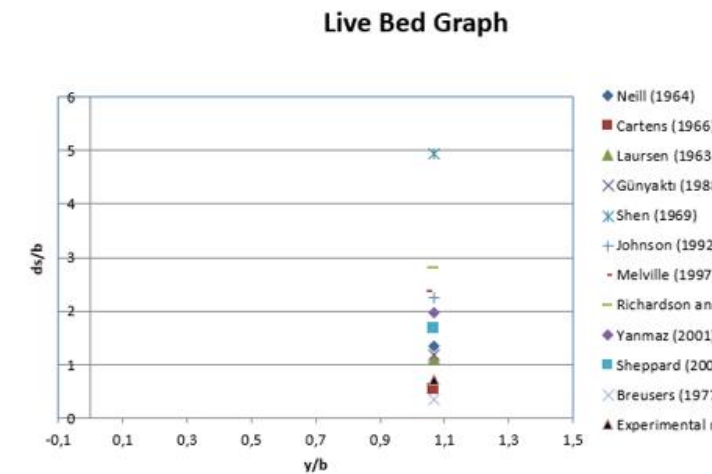
RESULTS AND DISCUSSION



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